

Chiara Di Vece, PhD

AI Lead · Morph

London, United Kingdom

✉ chiaradivece@gmail.com | 🌐 chiaradivece | in chiaradivece

Experience

Industry

AI Lead and Founding Applied Scientist

Morph Oct 2024 — Present
London, UK

- Driving the development of AI-based solutions across all the verticals with a focus on **Computer Vision, Agentic AI**, and **Foundation Models** for physical AI systems for soft robotics.
- Architecting and building the **Physical AI Platform**: an AI-driven co-design system for simulation-driven optimization of pneumatic soft robotic structures.
- Developing **computer vision software for robotic systems**, including vision-based perception stacks integrated with robot control software for autonomous operation.

AI Researcher

AI SuitUp (Remote) Jan 2023 — Jan 2025
Albuquerque, NM, USA

- Contributed to research and prototyping of AI-driven tools for automated **headshot generation**, enabling users to create professional profile images using generative AI.
- Optimized neural network pipelines for face detection, alignment, and style transfer, improving generation quality across diverse demographics.
- Benchmarked model performance and conducted experiments to reduce inference time, enabling scalable deployment for web-based services.
- Collaborated cross-functionally to evaluate aesthetic quality metrics and guide iterative model refinement based on user feedback.

Software Engineer Intern, Deep Learning Algorithms

MathWorks Jun 2023 — Sep 2023
Cambridge, UK

- Extended the Deep Learning Toolbox with the **sort method**, with full support in the Automatic Differentiation engine, enabling users to build custom differentiable layers with sorting operations.

Academia

Teaching Assistant

University College London, Department of Computer Science Sep 2021 — Jan 2024
London, UK
Running the coding lab sessions, Q&A sessions, and homework/exams development and marking for the following modules:

- **COMP0089: Reinforcement Learning (Google DeepMind)** - Explores how to implement, apply, and test relevant learning algorithms.
- **COMP0037: Robotic Systems** - Explores how intelligent system knowledge representation is translated into actions to achieve goals using reinforcement learning, both in pre-programmed and dynamic situations.
- **COMP0088: Introduction to Machine Learning** - Provides a grounding in the main techniques of "classical" machine learning, together with a briefer introduction to more advanced techniques.
- **ENGF0003: Mathematical Modelling and Analysis 1** - Helps to develop your understanding of core mathematical concepts and modelling and analysis skills that underpin the practice of engineering.

Research Intern

Politecnico di Milano, Department of Electronics, Information and Bioengineering Mar 2020 — Sep 2020
Milan, Italy

- Developed a **Unity**-based realistic and time-bounded simulator using a **novel approach** (position-based dynamics), validated against in-vivo animal trials (**mean mismatch: $4.73 \pm 2.15\%$**)
- Paper accepted for **publication** in **IEEE RA-L** and presented at **IEEE/RSJ IROS**
- MSc student supervision on thesis development (**KUKA robot and haptic interfaces** for palpation procedures)

Graduate Teaching Assistant

University of Illinois at Chicago, Department of Bioengineering Jan 2019 — May 2019
Chicago, IL, USA

- Running exercise practice sessions, Q&A sessions, and homework/exam marking.
- **BME310: Biological Systems Analysis** - System dynamics and frequency-domain analysis in bioengineering systems. Topics include **population models, predator-prey models, metabolic networks, biological oscillation, dynamics of infectious diseases**.

Graduate Research Student

University of Illinois at Chicago, Department of Urology & UIC Innovation Center

Jan 2019 – May 2019

Chicago, IL, USA

- Development of a novel **haptic and virtual reality-based simulator** for improvement of psychomotor skill in Veress needle insertion; the **System Usability Scale** test with experienced clinicians resulted in **“good” usability**.
- Paper accepted for **publication** in a health informatics journal (**ICARS**).  GitHub repo

Education

PhD in Computer Science, University College London

Licence to lawfully engage in the professional practice of Industrial Engineer, Italy

Master of Science, Biomedical Engineering, Politecnico di Milano, GPA: 110/110 *cum Laude*

Master of Science, Bioengineering, University of Illinois at Chicago, GPA: 4.00/4.00 | Double Degree

Bachelor of Science, Biomedical Engineering, Politecnico di Milano, GPA: 107/110

Professional Service

Reviewer for Journals and Conferences

- **Journals:** IEEE Transactions on Medical Imaging, Medical Image Analysis, IEEE Access, IEEE Robotics and Automation Letters, International Journal of Computer Assisted Radiology and Surgery.
- **Conferences:** IPCAI (*Outstanding Reviewer Award*), MICCAI, IEEE/RSJ IROS, IEEE ICRA, IEEE/CVF CVPR, ECCV, IEEE EMBC.

IEEE Women in Engineering Student Branch Affinity Group

Politecnico di Milano

Mar 2018 – Oct 2020

Milan, Italy

- Executive board member, social media pages manager, and networking events organizer.

Honors & Awards

2025 **Top 3 project for the Women's health tech grant 2024** | London, UK

2024 **Outstanding Reviewer Award** - recognized in the top 7% of reviewers for contributions to maintaining the scientific quality of the conference for the International Conference on Information Processing in Computer-Assisted Interventions | Barcelona, Spain; **Women's health tech grant**, UCL's EPSRC Impact Acceleration Account (£5000) | London, UK

2022 **AURIS/PCAI Young Investigator Travel Award** | **1st place** | Tokyo, Japan; **Invited journal paper in IJCARS** from best papers of IPCAI 2022 | Tokyo, Japan; **Microsoft/MathWorks WiMIUA Best Poster Award** | **2nd place** | Cambridge, UK; **MICCAI2022 Women in MICCAI Inspirational Leadership Legacy** | **1st place** | Singapore

2021 **Best group project at the UCL Medical Image Computing Summer School (MedICSS)** | **1st place** | London, UK

2020 **Alta Scuola Politecnica Top 3 thesis projects** | Turin, Italy; **UCL Research Studentship in the Department of Computer Science** (until Jan 2025) | London, UK

2019 **M.S. with Honors** | Politecnico di Milano and Politecnico di Torino | Milan, Italy; **Rare Disease Hackathon** | **3rd place** | Florence, Italy

2018 **IEEE Computer Society, Lance Stafford Larson Writing Award, Undergraduate Competition** | **3rd place**

2017 **Alta Scuola Politecnica** | selected among the top 1% students at Politecnico di Milano and Politecnico di Torino | Milan, Italy; **Rare Disease Hackathon** | **1st place** | Florence, Italy; **Young Professional Demo Competition, IEEE RTSI** | **1st place** | Modena, Italy

Invited Talks

FETUS: Freehand Echographic Training for Ultrasound System

Invited Presenter, Innovation in Women's Health Tech 2025
Impact Hub London

Mar 2025

London, UK

Deep Reinforcement Learning-based Guidance in Freehand Obstetric Ultrasound Training

Invited Presenter, Women in Medical Image Understanding and Analysis workshop (WiMIUA)
University of Cambridge

Jul 2022

Cambridge, UK

Embodiment of AI: How robots can interpret environment information and act

Lecturer and Organizer, PhD Spring School
Università della Calabria, Department of Computer Science

May 2020

Catanzaro, Italy

Theralight: Lightwave Technologies for Theranostic of Tumors

Invited Speaker, SharedCity: Innovation, Participation and Sustainability in Cities of the Future

Feb 2020

Milan, Italy

Publications (110+ citations)

1. **Di Vece, C.**, and Cirigliano, A., Le Lous, M., Napolitano, R., David, A.L., Peebles, D., and Jannin, P., Vasconcelos, F., Stoyanov, D. Measuring proximity to standard planes during fetal brain ultrasound scanning. **Under review - IEEE Transactions on Medical Imaging** (2026).
2. Saikia, A., **Di Vece, C.**, Mao, Z., Bonilla, S., He, C., Ramalhinho, J., Czempiel, T., Bano, S. and Stoyanov, D. HyKey: hyperspectral keypoint detection and matching in minimally invasive surgery. *International Journal of Computer Assisted Radiology and Surgery*, 1–9 (2026).
3. **Di Vece, C.**, Mao, Z., Avidris, N., Dromey, B., Napolitano, R., Ben Bashat, D., Vasconcelos, F., Stoyanov, D., Joskowicz, L. and Bano, S. A multicentre benchmark dataset for comprehensive landmark-based fetal ultrasound biometry. *Scientific Reports* (2026).
4. Saikia, A., **Di Vece, C.**, Mao, Z., Bonilla, S., He, C., Remalhinho, J., Czempiel, T., Bano, S. and Stoyanov, D. HyKey: Hyperspectral Keypoint Detection and Matching in Minimally Invasive Surgery in. Accepted for presentation at the 17th International Conference on Information Processing in Computer-Assisted Interventions (IPCAI 2026) and for publication in the International Journal of Computer Assisted Radiology and Surgery (IJCARs) (2025).
5. Saikia, A., **Di Vece, C.**, Bonilla, S., He, C., Magbagbeola, M., Mennillo, L., Czempiel, T., Bano, S. and Stoyanov, D. Robotic Arm Platform for Multi-View Image Acquisition and 3D Reconstruction in Minimally Invasive Surgery. *IEEE Robotics and Automation Letters* **10**, 3174–3181 (2025).
6. Bonilla*, S., **Di Vece*, C.**, Daher*, R., Ju*, X., Stoyanov, D., Vasconcelos, F. and Bano, S. *Mismatched: Evaluating the Limits of Image Matching Approaches and Benchmarks in Computer Vision – ECCV 2024 Workshops* ***Equal contribution** (2025), 120–137.
7. Le Lous, M., Vasconcelos, F., **Di Vece, C.**, Dromey, B., Napolitano, R., Yoo, S. and Edwards, E., Huulme, A., Peebles, D., Stoyanov, D. and Jannin, P. Probe motion during mid-trimester fetal anomaly scan in the clinical setting: A prospective observational study. *European Journal of Obstetrics & Gynecology and Reproductive Biology* **298**, 13–17 (2024).
8. **Di Vece, C.**, Le Lous, M., Dromey, B., Vasconcelos, F., David, A.L., Peebles, D. and Stoyanov, D. Ultrasound plane pose regression: assessing generalized pose coordinates in the fetal brain. *IEEE Transactions on Medical Robotics and Bionics* (2023).
9. **Di Vece, C.**, Dromey, B., Vasconcelos, F., David, A.L., Peebles, D. and Stoyanov, D. *Deep Learning-based Plane Pose Regression towards Training in Freehand Obstetric Ultrasound in Hamlyn Symposium on Medical Robotics* (2022), 49–50.
10. **Di Vece, C.**, Dromey, B., Vasconcelos, F., David, A.L., Peebles, D. and Stoyanov, D. Deep Learning-based Plane Pose Regression in Obstetric Ultrasound. *International Journal for Computer Assisted Radiology and Surgery (IJCARs)*. Accepted for presentation at the 13th International Conference on Information Processing in Computer-Assisted Interventions (IPCAI 2022) and awarded AURIS/IPCAI Young Investigator Travel Award (**1st place**) (2022).
11. Segato*, A., **Di Vece*, C.**, Zucchelli, S., Di Marzo, M., Wendler, T. Azampour, M.F., Galvan, S., Secoli, R. and De Momi, E. Position-Based Dynamics Simulator of Brain Deformations for Path Planning and Intra-Operative Control in Keyhole Neurosurgery. *IEEE Robotics and Automation Letters*. ***Equal contribution**; Accepted for presentation at the 2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2021) (2021).
12. **Di Vece, C.**, Luciano, C. and De Momi, E. Improvement of Psychomotor Skills Development for Veress Needle Placement using Haptics and Virtual Reality. *International Journal for Computer Assisted Radiology and Surgery (IJCARs)* (2021).
13. Crocioni*, G., **Di Vece*, C.** and Esmailbeigi, H. GLOS: GLOVe for Speech Recognition in 2019 41st Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC) ***Equal contribution** (2019), 3319–3322.
14. Bracco*, F., **Di Vece*, C.**, Cerina, L. and Santambrogio, M.D. BIE-PlnCS: Brain injury evaluation with pupillometer based on infrared camera system in 2017 IEEE 3rd International Forum on Research and Technologies for Society and Industry (RTSI) ***Equal contribution** (2017).

Advising & Mentoring

Research Mentor, Lumiere Education

2022 – Present

- Mentoring high school and undergraduate students globally through research programs founded by Harvard and Oxford PhDs, guiding students from research question development through to publication. Multiple mentees have published in peer-reviewed venues, including the National High School Journal of Science and the International Conference on Human-Computer Interaction.

Graduate Research Advisor, University College London & Politecnico di Milano

2020 – 2024

- Supervised MSc and visiting research students across projects spanning ultrasound pose estimation, haptic simulation, computer vision pipelines for clinical imaging, and surgical teleoperation systems

Extracurricular Experiences

Alta Scuola Politecnica (ASP), Technology Talents, Politecnico di Milano

Oct 2017 – Feb 2020

- Joint honor program of Politecnico di Milano and Politecnico di Torino running in parallel with the MSc and based on merit (top 1%, 150 participants per year) with courses in innovation, management, decision-making, soft skills, design methods and processes.
- Awarded with the ASP diploma and Double Degree in Biomedical Engineering from Politecnico di Torino (MSc), both *cum Laude*